THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50 and frequencies 8, 16, p, 10, 8 respectively is 23.8. The value of p is: 1

(a) 7 **(b)** 10 **(c)** 8 **(d)** 13

2. In the formula $\bar{x} = a + h\dfrac{\sum f_i u_i}{\sum f_i}$ (step-deviation method), u_i equals: 1

(a) $h(x_i - a)$ **(b)** $x_i - a$ **(c)** $\dfrac{x_i - a}{h}$ **(d)** $\dfrac{x_i}{h}$

3. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 13, 5, 20, 10, 2 respectively, The mean of the data is: 1

(a) 22.6 **(b)** 216 **(c)** 25 **(d)** 21.6

4. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 11, 10, 13, 6 respectively, The median of the data is: 1

(a) 33.8 **(b)** 34 **(c)** 30 **(d)** 40

5. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50 with frequencies 2, 11, 8, 17, 2 respectively, The mode of the data is: 1

(a) 30 **(b)** 33.75 **(c)** 40 **(d)** 26.5

6. For the data with classes 30–40, 40–50, 50–60, 60–70, 70–80 with frequencies 2, 11, 8, 17, 2 respectively, The modal class is: 1

(a) 60–70 **(b)** 50–60 **(c)** 40–50 **(d)** 30–40

7. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50 and frequencies 13, 5, 20, p, 2 respectively is 21.6. The value of p is: 1

(a) 9 **(b)** 15 **(c)** 10 **(d)** 12

8. The median of grouped data is found using: 1

(a) class marks only **(b)** the highest frequency only
(c) cumulative frequencies **(d)** deviations from the mean

9. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 12, 18, 11, 20, 21, 18 respectively, The mean of the data is: 1

(a) 82.4 **(b)** 85 **(c)** 83.4 **(d)** 81.4

10. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 5, 7, 5, 4, 4 respectively, The median of the data is: 1

(a) 41 **(b)** 50 **(c)** 43 **(d)** 40

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The daily electricity consumption (in units) of households in a colony is given below. 2

Units consumed	50–60	60–70	70–80	80–90	90–100	100–110
Number of households	8	10	14	2	13	3

Find the mode.

12. The distances (in km) travelled to office by employees of a company are given below. 2

Distance (km)	40–50	50–60	60–70	70–80	80–90	90–100
Number of employees	9	21	18	20	18	14

Find the median.

13. The ages (in years) of patients admitted to a hospital in a month are given below. **2**

Age (years)	20–30	30–40	40–50	50–60	60–70
Number of patients	4	13	4	16	3

Find the mean.

14. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. **2**

Lifetime (hours)	40–50	50–60	60–70	70–80	80–90
Number of bulbs	5	19	5	5	16

Find the mode.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The ages (in years) of patients admitted to a hospital in a month are given below. **3**

Age (years)	20–30	30–40	40–50	50–60	60–70
Number of patients	4	13	4	16	3

Find the mean.

16. The daily electricity consumption (in units) of households in a colony is given below. **3**

Units consumed	10–20	20–30	30–40	40–50	50–60
Number of households	2	3	7	10	3

Find the median.

17. The distances (in km) travelled to office by employees of a company are given below. **3**

Distance (km)	0–10	10–20	20–30	30–40	40–50
Number of employees	13	5	20	10	2

Find the mode.

OR

The weights (in kg) of students of a class are given below. **3**

Weight (kg)	60–70	70–80	80–90	90–100	100–110
Number of students	5	14	8	2	11

Find the mean.

18. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. **3**

Lifetime (hours)	60–70	70–80	80–90	90–100	100–110
Number of bulbs	2	11	10	8	9

Find the median.

OR

The monthly pocket money (in ₹) of students is given below. **3**

Pocket money (₹)	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	12	18	11	20	21	18

Find the mode.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 10, p, 10, 13, 6 respectively is 33.8. Find the value of p. **5**

OR

The median of a distribution with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 12, 6, x, 8, y, 4 respectively, and total frequency 50, is 83.75. Find the value of x. **5**

20. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 5, 7, p, 4, 4 respectively is 43. Find the value of p. **5**

OR

The median of a distribution with classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 5, 15, x, 16, y, 12, 15 respectively, and total frequency 100, is 67.5. Find the value of x. **5**

END OF QUESTION PAPER — Total: 40 Marks